Advanced Mathematics (Part 1) — Practice Problems: Detailed Solutions

1. Limit Computation and Proofs

1 Limits of Functions and Sequences

1

$$\lim_{n\to\infty}\left(\frac{1}{2}+\frac{3}{2^2}+\cdots+\frac{2n-1}{2^n}\right)$$

Solution Let

$$S_n=\frac{1}{2}+\frac{3}{2^2}+\cdots+\frac{2n-1}{2^n}$$

Use the shift-and-subtract method: multiply by the common ratio $\frac{1}{2}$ and subtract, which gives

$$S_n=3-\frac{2n+3}{2^n}$$

As $n\to\infty$, $\frac{2n+3}{2^n}\to 0$, so the limit is 3.

2

$$\lim_{n\to\infty}\left(\sqrt{n+2}-2\sqrt{n+1}+\sqrt{n}\right)$$

Solution Write it in difference form

$$\sqrt{n+2}-\sqrt{n+1}-\left(\sqrt{n+1}-\sqrt{n}\right)$$

After rationalizing we get

$$\frac{1}{\sqrt{n+2}+\sqrt{n+1}}-\frac{1}{\sqrt{n+1}+\sqrt{n}}$$

After combining over a common denominator, the numerator is $\sqrt{n}-\sqrt{n+2}$ and the denominator is a product of positive terms; using the Lagrange mean value theorem or a Taylor expansion, the limit is 0.

3

$$\lim_{x\to 1}\frac{\sqrt{x+8}-3}{\sqrt{x+3}-2}$$

Solution Rationalize both the numerator and the denominator. The numerator

$$\sqrt{x+8}-3=\frac{x-1}{\sqrt{x+8}+3}$$

The denominator

$$\sqrt{x+3}-2=\frac{x-1}{\sqrt{x+3}+2}$$

After cancelling $x-1$ we get

$$\frac{\sqrt{x+3}+2}{\sqrt{x+8}+3}\to\frac{2+2}{3+3}=\frac{2}{3}$$

The limit is $\dfrac{2}{3}$.

4

$$\lim_{x\to+\infty}\frac{(x-1)^2+1-3x}{x^2+2x^3}$$

Solution Expanding the numerator

$$(x-1)^2+1-3x=x^2-5x+2$$

The denominator is

$$2x^3+x^2$$

Dividing by the highest power x^3:

$$\frac{\dfrac{1}{x}-\dfrac{5}{x^2}+\dfrac{2}{x^3}}{2+\dfrac{1}{x}}\to\frac{0}{2}=0$$

The limit is 0.

5

$$\lim_{x\to-\infty}\frac{\sqrt{x^2-1}}{x}$$

Solution For $x<0$,

$$\sqrt{x^2-1}=|x|\sqrt{1-\frac{1}{x^2}}=-x\sqrt{1-\frac{1}{x^2}}$$

so

$$\frac{\sqrt{x^2-1}}{x}=-\sqrt{1-\frac{1}{x^2}}\to-1$$

The limit is -1.

6

$$\lim_{x\to0}\frac{\sin x^2}{1-\cos x}$$

Solution Equivalent-infinitesimal substitution

$$\sin x^2\sim x^2$$

$$1-\cos x \sim \frac{1}{2}x^2$$

so

$$\frac{x^2}{\frac{1}{2}x^2}=2$$

The limit is 2.

7

$$\lim_{x\to 0}\frac{\sin x-\tan x}{x^3}$$

Solution

$$\tan x=\frac{\sin x}{\cos x}$$

so

$$\sin x-\tan x=\sin x\left(1-\frac{1}{\cos x}\right)=\frac{\sin x(\cos x-1)}{\cos x}$$

Substituting into the limit

$$\lim_{x\to 0}\frac{\sin x(\cos x-1)}{x^3\cos x}=\lim_{x\to 0}\frac{x\cdot\left(-\frac{1}{2}x^2\right)}{x^3\cdot 1}=-\frac{1}{2}$$

The limit is $-\dfrac{1}{2}$.

8

$$\lim_{x\to 0}\left(1+2\tan^2 x\right)^{\cot^2 x}$$

Solution Let

$$y=\left(1+2\tan^2 x\right)^{\cot^2 x}$$

then

$$\ln y=\cot^2 x\cdot\ln\left(1+2\tan^2 x\right)$$

Let $t=\tan^2 x\to 0$; then

$$\cot^2 x=\frac{1}{t}$$

$$\ln y=\frac{\ln(1+2t)}{t}\to\frac{2t}{t}=2$$

so

$$y\to e^2$$

The limit is e^2.

9

$$\lim_{x \to 0} \frac{\tan x - \sin x}{\sin^3 x}$$

Solution

$$\tan x - \sin x = \sin x \left(\frac{1}{\cos x} - 1 \right) = \frac{\sin x (1 - \cos x)}{\cos x}$$

so

$$\frac{\tan x - \sin x}{\sin^3 x} = \frac{1 - \cos x}{\cos x \sin^2 x} \sim \frac{\frac{1}{2} x^2}{1 \cdot x^2} = \frac{1}{2}$$

The limit is $\frac{1}{2}$.

10

$$\lim_{x \to 0^+} \frac{\int_0^x t \sin t \, dt}{x^2 \ln(1 + x)}$$

Solution L'Hôpital's rule. Differentiating the numerator gives

$$x \sin x$$

Differentiating the denominator gives

$$2x \ln(1 + x) + \frac{x^2}{1 + x}$$

so

$$\lim_{x \to 0^+} \frac{x \sin x}{2x \ln(1 + x) + \frac{x^2}{1 + x}} = \lim_{x \to 0^+} \frac{\sin x}{2 \ln(1 + x) + \frac{x}{1 + x}}$$

By equivalent infinitesimals, $\sin x \sim x$ and $\ln(1 + x) \sim x$, and the denominator $\sim 2x + x = 3x$; the limit is $\frac{1}{3}$.

11

$$\lim_{x \to 0} \frac{\int_0^x \ln(1 + t) \, dt}{x \left(e^x - 1 \right)}$$

Solution L'Hôpital's rule. Differentiating the numerator gives

$$\ln(1 + x)$$

Differentiating the denominator gives

$$e^x - 1 + x e^x$$

so

$$\lim_{x \to 0} \frac{\ln(1 + x)}{e^x - 1 + x e^x}$$

By equivalent infinitesimals, $\ln(1 + x) \sim x$ and the denominator $\sim x + x = 2x$; the limit is $\frac{1}{2}$.

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12 Find the following limits

1

$$\lim_{x\to\pi}\frac{1+\cos x}{\tan^2 x}$$

Solution Let $t=x-\pi$; then

$$\cos x=-\cos t$$

$$\tan x=\tan t$$

The original expression becomes

$$\frac{1-\cos t}{\tan^2 t}\sim\frac{\frac{1}{2}t^2}{t^2}=\frac{1}{2}$$

The limit is $\frac{1}{2}$.

2

$$\lim_{x\to0}\frac{\tan x-x}{x^2\tan x}$$

Solution Taylor expansion

$$\tan x=x+\frac{1}{3}x^3+o(x^3)$$

so

$$\tan x-x\sim\frac{1}{3}x^3$$

$$x^2\tan x\sim x^3$$

The limit is $\frac{1}{3}$.

3

$$\lim_{x\to1}x^{\frac{1}{1-x}}$$

Solution Let $t=1-x$; then

$$x^{\frac{1}{1-x}}=(1-t)^{\frac{1}{t}}=\left[(1-t)^{-\frac{1}{t}}\right]^{-1}\to e^{-1}$$

The limit is $\frac{1}{e}$.

4

$$\lim_{x\to0^+}x^{\sin x}$$

Solution Take logarithms

$$\ln y=\sin x\ln x\sim x\ln x\to0$$

so

$$y\to1$$

The limit is 1.

5

$$\lim_{x\to 0}\frac{1-x^2-e^{-x^2}}{x^{\sin^3 2x}}$$

Solution The denominator

$$x^{\sin^3 2x}$$

Taking the logarithm gives

$$\sin^3 2x\ln x\sim 8x^3\ln x\to 0$$

so the denominator tends to 1. Taylor-expanding the numerator

$$e^{-x^2}=1-x^2+\frac{1}{2}x^4+o(x^4)$$

therefore

$$1-x^2-e^{-x^2}=-\frac{1}{2}x^4+o(x^4)$$

The limit is 0.

6

$$\lim_{x\to 0}\left(\frac{1}{x}-\frac{\cos x}{\sin x}\right)\cdot\frac{1}{x}$$

Solution Combine over a common denominator

$$\frac{1}{x}-\frac{\cos x}{\sin x}=\frac{\sin x-x\cos x}{x\sin x}$$

The original expression

$$=\frac{\sin x-x\cos x}{x^2\sin x}$$

Taylor expansion

$$\sin x=x-\frac{1}{6}x^3$$

$$\cos x=1-\frac{1}{2}x^2$$

then

$$\sin x-x\cos x=\frac{1}{3}x^3+o(x^3)$$

so the limit is $\frac{1}{3}$.

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2 Proof Problems

1

$f(x)$ is continuous on $[0,2]$ and $f(0)=f(2)$; prove that there exists at least one point $\xi \in [0,1]$ such that $f(\xi)=f(\xi+1)$.

Proof Construct the auxiliary function

$$F(x)=f(x)-f(x+1)$$

$$x \in [0,1]$$

then

$$F(0)=f(0)-f(1)$$

$$F(1)=f(1)-f(2)=f(1)-f(0)=-F(0)$$

If $F(0)=0$, then $\xi=0$ works; otherwise $F(0)$ and $F(1)$ have opposite signs, so by the zero-point theorem there exists $\xi \in (0,1)$ such that

$$F(\xi)=0$$

that is,

$$f(\xi)=f(\xi+1)$$

---

2 Proof: when

$x \geq 1$, we have

$$2\arctan x+\arcsin\frac{2x}{1+x^2}=\pi$$

Proof Let

$$f(x)=2\arctan x+\arcsin\frac{2x}{1+x^2}$$

Differentiating gives

$$f'(x)=0$$

so $f(x)$ is constant. Computing

$$f(1)=2\cdot\frac{\pi}{4}+\arcsin 1=\frac{\pi}{2}+\frac{\pi}{2}=\pi$$

therefore

$$f(x)\equiv\pi$$

---

3 Suppose the function

$f(x)$ is continuous on the closed interval $[0,1]$ and differentiable on the open interval $(0,1)$, with $f(1)=0$. Prove that in the open interval $(0,1)$ there exists at least one point ξ such that

$$f'(\xi)=-\frac{1}{\xi}f(\xi)$$

Proof Construct

$$F(x) = xf(x)$$

then

$$F(0) = 0$$

$$F(1) = 0$$

By Rolle's theorem, there exists $\xi \in (0, 1)$ such that

$$F'(\xi) = 0$$

that is,

$$\xi f'(\xi) + f(\xi) = 0$$

so

$$f'(\xi) = -\frac{1}{\xi} f(\xi)$$

---

4 Proof

$$pb^{p-1}(a-b) \leq a^p - b^p \leq pa^{p-1}(a-b)$$

where $0 < b < a$ and $p > 1$.

Proof Apply the Lagrange mean value theorem to $f(x) = x^p$ on $[b, a]$:

$$a^p - b^p = p\xi^{p-1}(a-b)$$

$$b < \xi < a$$

Since $p > 1$, x^{p-1} is increasing, so

$$b^{p-1} < \xi^{p-1} < a^{p-1}$$

Multiplying by the positive number $p(a-b)$ gives the inequality.

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2. Derivatives and Differentials

1 For the curve

$$\begin{cases} x = te^{-t} \\ y = e^t \end{cases}$$

find the equations of the tangent line and the normal line at the point where $t = 0$.

Solution When $t = 0$, $x = 0$ and $y = 1$.

$$\frac{dx}{dt} = (1-t)e^{-t}$$

$$\frac{dy}{dt} = e^t$$

$$\frac{dy}{dx} = \frac{e^t}{(1-t)e^{-t}} = \frac{e^{2t}}{1-t}$$

At $t = 0$ the slope is 1. The tangent line is

$$y = x + 1$$

The normal line is

$$y = -x + 1$$

---

2 For the curve

$$y = \ln x$$

find the equation of the tangent line on it that is parallel to the line $y = x$.

Solution

$$y' = \frac{1}{x}$$

Setting $\dfrac{1}{x} = 1$ gives $x = 1$, so the point of tangency is $(1, 0)$, and the tangent line is

$$y = x - 1$$

---

3 Suppose

$$y = e^x \ln x$$

find y'.

Solution By the product rule

$$y' = e^x \ln x + e^x \cdot \frac{1}{x} = e^x \left(\ln x + \frac{1}{x} \right)$$

---

4 Suppose

$$y = \left(\arctan \frac{x}{3} \right)^2$$

find y'.

Solution By the chain rule

$$y' = 2 \arctan \frac{x}{3} \cdot \frac{1}{1 + \left(\frac{x}{3}\right)^2} \cdot \frac{1}{3} = \frac{6 \arctan \frac{x}{3}}{9 + x^2}$$

---

5 Suppose

$$y = \sin^2 x \cdot \sin(x^2)$$

find y'.

Solution By the product rule

$$y' = \sin 2x \cdot \sin(x^2) + \sin^2 x \cdot 2x \cos(x^2)$$

---

6 Suppose

$$y = \log_2(e^x - 1)$$

find y'.

Solution Change of base

$$y = \frac{\ln(e^x - 1)}{\ln 2}$$
$$y' = \frac{1}{\ln 2} \cdot \frac{e^x}{e^x - 1}$$

---

7 For the astroid

$$x^3 + y^3 = a^3 \ (a > 0)$$

find the equation of the tangent line at the point $M\left(\frac{\sqrt{2}}{4}a, \frac{\sqrt{2}}{4}a\right)$.

Solution Implicit differentiation

$$3x^2 + 3y^2 y' = 0 \Rightarrow y' = -\frac{x^2}{y^2}$$

At M we have $x = y$, so

$$y' = -1$$

The tangent line is

$$x + y = \frac{\sqrt{2}}{2}a$$

---

8 Suppose the function

$$y = \operatorname{arctan} \frac{1}{x}$$

find dy and $dy\big|_{x=1}$.

Solution

$$y' = -\frac{1}{1 + x^2}$$

so

$$dy = -\frac{1}{1 + x^2}dx$$
$$dy\big|_{x=1} = -\frac{1}{2}dx$$

---

9 Suppose

$$y = y(x)$$

is determined by the equation

$$2xy = x + y$$

find $dy\big|_{x=0}$.

Solution Differentiate both sides

$$2(y\,dx + x\,dy) = dx + dy$$

Rearranging gives

$$(2x - 1)dy = (1 - 2y)dx$$

so

$$dy = \frac{1 - 2y}{2x - 1}dx$$

When $x = 0$, the original equation gives $y = 0$, so

$$dy\big|_{x=0} = -dx$$

---

10 Given

$$y = \ln^2(1 - x)$$

find dy.

Solution

$$y' = -\frac{2\ln(1 - x)}{1 - x}$$

$$dy = -\frac{2\ln(1 - x)}{1 - x}dx$$

---

11 Given

$$y = \int_0^x \tan t^2\,dt$$

find y'.

Solution By the Fundamental Theorem of Calculus

$$y' = \tan x^2$$

---

3. Indefinite and Definite Integrals

1 Computing Indefinite Integrals

1

$$\int \left(1-\sqrt[3]{x^2}\right)^2 dx$$

Solution

$$\sqrt[3]{x^2} = x^{2/3}$$

Expanding

$$\left(1-x^{2/3}\right)^2 = 1 - 2x^{2/3} + x^{4/3}$$

Integrating gives

$$x - \frac{6}{5}x^{5/3} + \frac{3}{7}x^{7/3} + C$$

2

$$\int \frac{1}{\cos^2 x \sin^2 x} dx$$

Solution

$$\frac{1}{\cos^2 x \sin^2 x} = \sec^2 x + \csc^2 x$$

Integrating gives

$$\tan x - \cot x + C$$

3

$$\int \frac{1}{4x^2 + a^2} dx$$

Solution

$$\int \frac{1}{4x^2 + a^2} dx = \frac{1}{2a} \arctan \frac{2x}{a} + C$$

4

$$\int x^2 e^{-x^3} dx$$

Solution Let

$$u = -x^3$$

then

$$\int x^2 e^{-x^3}\,dx = -\frac{1}{3}e^{-x^3} + C$$

5

$$\int \frac{e^{2x}}{1+e^x}dx$$

Solution Let

$$u = e^x$$

then

$$\int \frac{e^{2x}}{1+e^x}dx = e^x - \ln(1+e^x) + C$$

6

$$\int \sqrt{a^2 - x^2}\,dx \quad (a > 0)$$

Solution Let

$$x = a\sin t$$

then

$$\int \sqrt{a^2 - x^2}\,dx = \frac{a^2}{2}\arcsin\frac{x}{a} + \frac{x}{2}\sqrt{a^2 - x^2} + C$$

7

$$\int xe^x dx$$

Solution Integration by parts

$$\int xe^x dx = e^x(x-1) + C$$

8

$$\int x\ln x\,dx$$

Solution Integration by parts

$$\int x\ln x\,dx = \frac{x^2}{2}\ln x - \frac{x^2}{4} + C$$

9

$$\int \sec^3 x\,dx$$

Solution Integration by parts gives

$$\int \sec^3 x\,dx = \frac{1}{2}\left(\sec x \tan x + \ln\left|\sec x + \tan x\right|\right) + C$$

---

2 Computing Definite Integrals

1

$$\int_0^{\pi} \sqrt{1+\cos 2x}\,dx$$

Solution

$$\sqrt{1+\cos 2x} = \sqrt{2}\left|\cos x\right|$$

Integrating piecewise gives

$$2\sqrt{2}$$

2

$$\int_0^{2\pi} \left|\sin x\right| dx$$

Solution Using symmetry

$$\int_0^{2\pi} \left|\sin x\right| dx = 2\int_0^{\pi} \sin x\,dx = 4$$

3

$$\int_{-1}^{3} \frac{dt}{1+t^2}$$

Solution

$$\int_{-1}^{3} \frac{dt}{1+t^2} = \arctan 3 + \frac{\pi}{4}$$

4

$$\int_e^{e} \left|\ln x\right| dx$$

Solution The interval of integration has length 0, so the integral is 0.

5

$$\int_0^{\pi^2} \sqrt{x}\cos\sqrt{x}\,dx$$

Solution Let

$$t = \sqrt{x}$$

then

$$\int_0^{\pi^2} \sqrt{x}\cos\sqrt{x}\,dx = 2\int_0^{\pi} t^2\cos t\,dt = -4\pi$$

6

$$\int_0^{\pi/2} e^x\cos x\,dx$$

Solution Integrating by parts twice gives

$$\int_0^{\pi/2} e^x\cos x\,dx = \frac{1}{2}\left(e^{\pi/2}-1\right)$$

---

4. Applications of Integration: Area, Volume, Arc Length

1 Region enclosed by the parabola

$$y^2 = 2x$$

and the line

$$y = x - 4$$

Compute the area of the enclosed region.

Solution The intersection points are $(2,-2)$ and $(8,4)$. Integrating with respect to y

$$S = \int_{-2}^{4}\left(y + 4 - \frac{y^2}{2}\right)dy = 18$$

---

2 Region enclosed by the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Compute the volume of the solid of revolution obtained by revolving the enclosed region one full turn about the x-axis.

Solution

$$V = \pi\int_{-a}^{a} y^2\,dx = \frac{4}{3}\pi ab^2$$

---

3 A plane passes through the center of the base circle

of a cylinder of base radius R, and makes an angle α with the base plane; compute the volume of the solid cut from the cylinder by this plane.

Solution Take the plane

$$z = x\tan\alpha$$

The solid lies in the part $x \geq 0$, and the cross-sectional area is

$$2\sqrt{R^2 - x^2}\cdot x\tan\alpha$$

Integrating gives

$$V = \frac{2}{3}R^3\tan\alpha$$

---

4 The cardioid

$$\rho = a(1+\cos\theta)\ (a>0)$$

Compute the area of the region it encloses.

Solution

$$A = \frac{1}{2}\int_0^{2\pi} a^2(1+\cos\theta)^2\,d\theta = \frac{3}{2}\pi a^2$$

---

5 The cardioid

$$\rho = a(1+\cos\theta)\ (a>0)$$

Find its arc length.

Solution

$$L = \int_0^{2\pi}\sqrt{\rho^2+(\rho')^2}\,d\theta = 8a$$

---

6 The equations of the cycloid are

$$\begin{cases} x = a(t-\sin t) \\ y = a(1-\cos t) \end{cases} \quad (a>0) \quad (0\leq t\leq 2\pi)$$

1 Find the area of the region enclosed by one arch of the cycloid and the

x-axis.

Solution

$$A = \int_0^{2\pi} yx'\,dt = a^2\int_0^{2\pi}(1-\cos t)^2\,dt = 3\pi a^2$$

2 For the region enclosed by one arch of the cycloid and the

x-axis, find the volumes of the solids obtained by revolving it one full turn about the x-axis and the y-axis, respectively.

Solution About the x-axis

$$V_x = \pi\int_0^{2\pi} y^2x'\,dt = 5\pi^2a^3$$

About the y-axis

$$V_y = 2\pi \int_0^{2\pi} xyx'\,dt = 6\pi^3 a^3$$

3 Find the length of one arch of the cycloid

Solution

$$L = \int_0^{2\pi} \sqrt{x'^2 + y'^2}\,dt = 8a$$

---

5. Taylor's Formula and the Behavior of Functions

1 For the function

$$f(x) = \sin x$$

find its nth-order Maclaurin formula with the Lagrange remainder.

Solution

$$\sin x = \sum_{k=0}^{m-1} (-1)^k \frac{x^{2k+1}}{(2k+1)!} + R_n(x)$$

where

$$R_n(x) = \frac{\sin^{(n+1)}(\theta x)}{(n+1)!} x^{n+1}$$

$$\theta \in (0,1)$$

---

2 For the function

$$f(x) = e^x$$

find its nth-order Maclaurin formula with the Lagrange remainder.

Solution

$$e^x = \sum_{k=0}^{n} \frac{x^k}{k!} + R_n(x)$$

$$R_n(x) = \frac{e^{\xi}}{(n+1)!} x^{n+1}$$

$$\xi \in (0, x)$$

---

3 For

$$y = \ln(1 + x^2)$$

find the intervals of concavity and the inflection points.

Solution

$$y' = \frac{2x}{1+x^2}$$

$$y'' = \frac{2(1-x^2)}{(1+x^2)^2}$$

Setting $y'' = 0$ gives

$$x = \pm 1$$

Concave interval

$$(-1, 1)$$

Convex interval

$$(-\infty, -1) \cup (1, +\infty)$$

Inflection points

$$(\pm 1, \ln 2)$$

---

6. Power Series

1 Find the radius of convergence, open interval of convergence, and interval of convergence of the following power series

1

$$\sum_{n=1}^{\infty} \frac{x^n}{2^n n}$$

Solution Radius of convergence

$$R = 2$$

Open interval of convergence

$$(-2, 2)$$

Interval of convergence

$$[-2, 2)$$

2

$$\sum_{n=1}^{\infty} (\sqrt{n+1} - \sqrt{n})(x-2)^n$$

Solution Radius of convergence

$$R = 1$$

Open interval of convergence

$$(1, 3)$$

Interval of convergence

$$[1, 3)$$

3

$$\sum_{n=1}^{\infty} \frac{3^n + (-2)^n}{n} x^n$$

Solution Radius of convergence

$$R = \frac{1}{3}$$

Open interval of convergence

$$\left(-\frac{1}{3}, \frac{1}{3}\right)$$

Interval of convergence

$$\left[-\frac{1}{3}, \frac{1}{3}\right)$$

---

2 For the power series

$$\sum_{n=1}^{\infty} \frac{x^{n+1}}{n}$$

find the sum function, and for the series

$$\sum_{n=1}^{\infty} \frac{1}{n \cdot 2^n}$$

find the sum.

Solution The sum function

$$S(x) = -x \ln(1 - x)$$

$$|x| < 1$$

Let

$$x = \frac{1}{2}$$

which gives

$$\sum_{n=1}^{\infty} \frac{1}{n \cdot 2^n} = \ln 2$$

---

3 For the power series

$$\sum_{n=1}^{\infty} n x^n$$

find the sum function, and use it for the numerical series

$$\sum_{n=1}^{\infty} \frac{n}{2^n}$$

to find its sum.

Solution The sum function

$$S(x) = \frac{x}{(1-x)^2}$$

$$|x| < 1$$

Let

$$x = \frac{1}{2}$$

which gives

$$\sum_{n=1}^{\infty} \frac{n}{2^n} = 2$$

---

4 For

$$\sum_{n=1}^{\infty} \frac{x^n}{n(n+1)}$$

find the sum function.

Solution Decompose

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$$

which gives the sum function

$$S(x) = 1 + \frac{(1-x)\ln(1-x)}{x}$$

$$|x| < 1,\ x \neq 0$$

$$S(0) = 0$$